Dan

Formal System of the Neurol Substrate

Full Algorithmic, Mathematical, and Operational Formalization

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Abstract

This document is the second major volume of the revised neurol series and carries the name *Dan*. If *Sinh* establishes the philosophical, ontological, and foundational legitimacy of the neurol substrate, then *Dan* provides its formal system. This document defines the state variables, spaces, operators, thresholds, update rules, cost functions, reward equations, semantic resonance metrics, emergence criteria, lineage transitions, and boundedness conditions required to make the substrate mathematically explicit. The goal is not to present one final canonical equation for every future implementation detail, but to establish a coherent and sufficiently complete formal language such that later implementation in *Chuyen* I and *Chuyen* II can proceed without conceptual ambiguity.

Contents

1 Introduction **5**
1.1 Role of *Dan* . . . 5
1.2 What This Volume Must Achieve . . . 5
1.3 Status of the Formal System . . . 5

2 Global Mathematical Setting **6**
2.1 Time Indexing . . . 6
2.2 The Substrate Universe . . . 6
2.3 The Neurol Population . . . 6
2.4 Signals, Edges, and Events . . . 6
2.5 State Spaces . . . 6

3 Structured Neurol State **6**
3.1 State Decomposition . . . 6
3.2 Identity Component . . . 7
3.3 Activation Component . . . 7
3.4 Drive Component . . . 7
3.5 Resource Component . . . 8
3.6 Memory Component . . . 8
3.7 Semantic Component . . . 8
3.8 Existence and Influence Components . . . 8
3.9 Mode Component . . . 8

4 Signal Formalization **9**
4.1 Signal Structure . . . 9
4.2 Signal Energy Decay . . . 9
4.3 Signal Contribution to Target State . . . 9

5 Drive Accumulation and *Dan* Dynamics **9**
5.1 Drive Update Rule . . . 9
5.2 Excitation Term . . . 10
5.3 Internal Tension Term . . . 10
5.4 Drive Dissipation . . . 10

6 Activation Potential and Threshold Crossing **10**
6.1 Activation Potential . . . 10
6.2 Dynamic Threshold . . . 10
6.3 Commit Condition . . . 11
6.4 Transition Commit Operator . . . 11

7 Event Construction and Causal Ledger **11**
7.1 Event Structure . . . 11
7.2 Causal Graph . . . 12
7.3 Replay Condition . . . 12

8 Memory Formalization **12**
8.1 Local Memory Update . . . 12
8.2 Memory Importance Score . . . 12
8.3 Memory Consolidation . . . 12
8.4 Latent Inhibition . . . 13

9 Identity Trajectory Formalization **13**
9.1 Identity as a Dynamical Variable . . . 13
9.2 A Linear-Nonlinear Hybrid Update . . . 13
9.3 Trajectory Distance . . . 13

10 Semantic Space and Dynamic Context Field **13**
10.1 Semantic Coordinates . . . 13
10.2 Anchor Set Construction . . . 14
10.3 Context Plane Construction . . . 14
10.4 Context-Dependent Semantic Distance . . . 14
10.5 Semantic Resonance . . . 14
10.6 Imagination Mapping . . . 14

11 Topology and Edge Dynamics **14**
11.1 Edge State . . . 14
11.2 Edge Strength Update . . . 15
11.3 Bridge Discovery . . . 15

12 Resource Physics **15**
12.1 Local Resource State . . . 15
12.2 Resource Update . . . 15

12.3 Activation Cost . . . 16
12.4 Signal Propagation Cost . . . 16
12.5 Entropy Tax . . . 16

13 Existence Quantum, Reward, and Cooperation **16**
13.1 Existence Quantum . . . 16
13.2 Existence Update Rule . . . 16
13.3 Impact Reward . . . 17
13.4 Cooperation Score . . . 17
13.5 Influence Update . . . 17

14 Attention Allocation **17**
14.1 Attention Priority Score . . . 17
14.2 Attention Allocation Distribution . . . 17
14.3 Effect of Attention . . . 18

15 Cluster Emergence and Agent Promotion **18**
15.1 Local Coherence Between Neurols . . . 18
15.2 Cluster Score . . . 18
15.3 Agent Promotion . . . 18

16 ARC Reproduction and Mutation **18**
16.1 Entity Representation for Reproduction . . . 18
16.2 Compression Operator . . . 19
16.3 Decompression Operator . . . 19
16.4 Anchor Inheritance . . . 19
16.5 Interaction Types . . . 19
16.6 Reproduction Eligibility . . . 19

17 Compression, Dormancy, and Quarantine Operators **20**
17.1 Compression Trigger . . . 20
17.2 Dormancy Trigger . . . 20
17.3 Quarantine Trigger . . . 20

18 Global Runtime Update Equations **20**
18.1 Canonical Neurol Update . . . 20
18.2 Post-Commit Global Pass . . . 20

19 Invariants and Boundedness Conditions **21**
19.1 Non-Negativity of Resource . . . 21
19.2 Finite Event Commitment . . . 21
19.3 Existence Preservation . . . 21
19.4 Replay Consistency . . . 21
19.5 Bounded Reward Growth . . . 21

20 Recommended Formal Parameter Families **22**
20.1 Core Parameter Groups . . . 22
20.2 Why Parameters Must Remain Explicit . . . 22

21 What Remains for *Chuyen* I and *Chuyen* II **22**

21.1 Why Formalization Is Not the End 22
21.2 What *Chuyen* I Must Do 22
21.3 What *Chuyen* II Must Do 22

22 Conclusion **22**

1. Introduction

1.1. Role of *Dan*

Dan is the formal continuation of *Sinh*. The first volume argued that the neurol substrate requires a precise ontology centered on local stateful cells, transformation rather than annihilation, resonance rather than arbitrary interaction, and identity as trajectory rather than snapshot. Those claims now require mathematical and algorithmic articulation.

The purpose of *Dan* is therefore to turn the conceptual commitments of *Sinh* into a formal system. Every important primitive introduced there must here receive at least one of the following:

a. a state definition;

b. an operator;

c. an update rule;

d. a constraint;

e. a score function;

f. a boundedness condition;

g. or a transition law.

1.2. What This Volume Must Achieve

This volume must make explicit the mathematical machinery for:

1. neurol state and local dynamics;
2. signal propagation and decay;
3. drive accumulation and threshold crossing;
4. event construction and causal trace generation;
5. memory consolidation and identity trajectory update;
6. semantic context and resonance;
7. reward, cooperation, existence, and influence;
8. resource accounting and entropy;
9. cluster and agent emergence;
10. ARC-style reproduction, mutation, and pruning.

1.3. Status of the Formal System

The present system is designed to be strong enough for implementation and experimentation, while remaining open to future refinement. Some functions are therefore intentionally parameterized rather than frozen to one rigid numerical choice. This is not a weakness. It is the correct degree of formal flexibility for a theory at this stage.

2. Global Mathematical Setting

2.1. Time Indexing

Let the substrate evolve over discrete event-indexed or scheduler-indexed time steps

$$t \in \mathbb{N} = \{0, 1, 2, \ldots\}.$$

This discrete indexing is not meant to deny the continuity of certain phenomena. It is chosen because the implementation substrate commits transitions as explicit events. Continuous variants may later be considered as limits.

2.2. The Substrate Universe

Let the substrate at time t be denoted

$$\mathscr{V}_t.$$

It contains all active, dormant, compressed, or otherwise ontologically valid structures at time t, together with their relations, traces, resources, and residues.

2.3. The Neurol Population

Let the set of neurols present at time t be

$$\mathscr{N}_t = \{n_1, n_2, \ldots, n_{N_t}\}.$$

The cardinality $N_t = |\mathscr{N}_t|$ may change through creation, compression, dormancy, reproduction, and decomposition.

2.4. Signals, Edges, and Events

Let:

$$\begin{aligned}
\mathscr{S}_t &= \text{set of live signals at time } t, \\
\mathscr{E}_t &= \text{set of edges or relations at time } t, \\
\mathscr{Q}_t &= \text{set of committed events up to time } t.
\end{aligned}$$

2.5. State Spaces

Each neurol n_i is associated with a state vector

$$x_i(t) \in \mathscr{X},$$

where $\mathscr{X}$ is the full neurol state space. Later we decompose this into structured components.

3. Structured Neurol State

3.1. State Decomposition

We define the state of neurol n_i at time t as a structured tuple

$$x_i(t) = \big(\iota_i, \alpha_i(t), \delta_i(t), \rho_i(t), \mu_i(t), \kappa_i(t), \xi_i(t), \eta_i(t), \pi_i(t), \lambda_i(t)\big).$$

Here:

Component	Meaning
ι_i	stable identity and lineage reference
$\alpha_i(t)$	activation state and readiness-related fields
$\delta_i(t)$	local drive or *Dan*-accumulation state
$\rho_i(t)$	resource state
$\mu_i(t)$	memory state
$\kappa_i(t)$	semantic/context state
$\xi_i(t)$	existence-viability state
$\eta_i(t)$	influence and cooperation state
$\pi_i(t)$	mode state (active, dormant, compressed, etc.)
$\lambda_i(t)$	local causal and ledger references

3.2. Identity Component

The identity component is

$$\iota_i = (\mathrm{id}_i, \mathrm{lineage}_i, \mathrm{type}_i, q_i, d_i),$$

where:

- id_i is a unique neurol identifier;
- $\mathrm{lineage}_i$ identifies ancestry or transformation lineage;
- type_i identifies interaction class or internal subtype;
- q_i is abstraction level or quantum step;
- d_i is domain class or domain label.

3.3. Activation Component

Let

$$\alpha_i(t) = (a_i(t), r_i(t), h_i(t), \tau_i(t)),$$

where:

- $a_i(t)$ is activation potential;
- $r_i(t)$ is readiness;
- $h_i(t)$ is inhibition or damping;
- $\tau_i(t)$ is refractory or stabilization state.

3.4. Drive Component

Let

$$\delta_i(t) = d_i^{\mathrm{acc}}(t)$$

be the accumulated drive, tension, or propagation pressure of the neurol.

3.5. Resource Component

Let

$$\rho_i(t) = (R_i(t), C_i^{\text{burn}}(t), B_i(t))$$

where:

- $R_i(t)$ is available resource stock;
- $C_i^{\text{burn}}(t)$ is recent cost expenditure;
- $B_i(t)$ is resource balance trend or buffer.

3.6. Memory Component

Let

$$\mu_i(t) = \left(M_i^{\text{loc}}(t), M_i^{\text{epi}}(t), M_i^{\text{cmp}}(t)\right),$$

where:

- $M_i^{\text{loc}}(t)$ is local active memory;
- $M_i^{\text{epi}}(t)$ is episodic trace memory;
- $M_i^{\text{cmp}}(t)$ is compressed memory state.

3.7. Semantic Component

Let

$$\kappa_i(t) = \left(v_i(t), A_i(t), \Phi_i(t)\right),$$

where:

- $v_i(t) \in \mathbb{R}^m$ is the semantic coordinate vector;
- $A_i(t)$ is the anchor set;
- $\Phi_i(t)$ is the currently active frame or context plane reference.

3.8. Existence and Influence Components

Let

$$\xi_i(t) = \Xi_i(t)$$

be the existence quantum or viability measure, and let

$$\eta_i(t) = \left(I_i(t), \Gamma_i(t)\right)$$

where $I_i(t)$ is influence and $\Gamma_i(t)$ is cooperation contribution.

3.9. Mode Component

The mode state is categorical:

$$\pi_i(t) \in \{\text{active}, \text{latent}, \text{refractory}, \text{dormant}, \text{compressed}, \text{quarantine}\}.$$

4. Signal Formalization

4.1. Signal Structure

A signal $s \in \mathscr{S}_t$ is defined as

$$s = (u, w, y, \varepsilon, \omega, \theta, \ell, \zeta),$$

where:

- u is source identifier;
- w is target identifier or routing policy;
- y is payload vector;
- $\varepsilon \in \mathbb{R}_{\geq 0}$ is signal energy;
- ω is signal weight;
- θ is signal type;
- ℓ is time-to-live or expiration count;
- ζ is causal parent reference.

4.2. Signal Energy Decay

If a signal persists for k routing steps, we may model energy decay as

$$\varepsilon_{k+1} = \varepsilon_k \exp(-\lambda_s) \cdot \chi_k,$$

where $\lambda_s > 0$ is a decay rate and $\chi_k \in (0, 1]$ is an attenuation factor depending on route quality, interference, or congestion.

4.3. Signal Contribution to Target State

Given a neurol n_i receiving a set of incoming signals $\mathscr{S}_i^{\text{in}}(t)$, define the aggregated incoming force

$$F_i(t) = \sum_{s \in \mathscr{S}_i^{\text{in}}(t)} \omega_s \varepsilon_s \, g_\theta(y_s, x_i(t)),$$

where g_θ is a type-dependent interaction function.

This allows different signals to affect different state subspaces. For instance, memory signals may update $\mu_i(t)$ more than $\alpha_i(t)$, while inhibitory signals may increase $h_i(t)$.

5. Drive Accumulation and *Dan* Dynamics

5.1. Drive Update Rule

The drive accumulation state evolves by a general update rule of the form

$$d_i^{\text{acc}}(t+1) = \beta_d d_i^{\text{acc}}(t) + \varphi_i(t) + \psi_i(t) - \omega_i(t),$$

where:

- $\beta_d \in [0,1]$ is drive persistence;
- $\varphi_i(t)$ is incoming excitation from signals;
- $\psi_i(t)$ is internally generated tension, need, unresolved memory, or goal pressure;
- $\omega_i(t)$ is dissipation or release.

5.2. Excitation Term

A natural choice is

$$\varphi_i(t) = \alpha_F \|F_i(t)\|,$$

with $\alpha_F > 0$.

5.3. Internal Tension Term

Internal tension may depend on unresolved memory, semantic contradiction, or unmet resource demand. A general form is

$$\psi_i(t) = \alpha_M U_i(t) + \alpha_K C_i^{\text{sem}}(t) + \alpha_R D_i^{\text{res}}(t),$$

where:

- $U_i(t)$ is unresolved local memory tension;
- $C_i^{\text{sem}}(t)$ is semantic contradiction or ambiguity;
- $D_i^{\text{res}}(t)$ is resource demand mismatch.

5.4. Drive Dissipation

After action, stabilization, or prolonged inactivity, drive dissipates. A simple form is

$$\omega_i(t) = \alpha_\tau \tau_i(t) + \alpha_{\text{idle}} \mathbb{1}_{\text{idle}}(i,t).$$

6. Activation Potential and Threshold Crossing

6.1. Activation Potential

Define activation potential by

$$a_i(t) = w_d d_i^{\text{acc}}(t) + w_f \|F_i(t)\| + w_m m_i^{\text{act}}(t) - w_h h_i(t) - w_\tau \tau_i(t),$$

where $m_i^{\text{act}}(t)$ is the contribution of local active memory to readiness.

6.2. Dynamic Threshold

Let the threshold for neurol i be

$$\Theta_i(t) = \Theta_i^{(0)} + \Theta_i^{\text{mode}}(t) + \Theta_i^{\text{res}}(t) + \Theta_i^{\text{ctx}}(t),$$

where:

- $\Theta_i^{(0)}$ is base threshold;

- $\Theta_i^{\text{mode}}(t)$ depends on mode and refractory state;
- $\Theta_i^{\text{res}}(t)$ penalizes low-resource situations;
- $\Theta_i^{\text{ctx}}(t)$ depends on semantic or contextual conditions.

6.3. Commit Condition

A *Chuyen* event is eligible when

$$a_i(t) \geq \Theta_i(t)$$

and the neurol is not in a disallowed mode such as compressed or quarantine.

Definition 6.1 (Transition Eligibility)**.** *A neurol n_i is transition-eligible at time t if and only if*

$$a_i(t) \geq \Theta_i(t), \quad R_i(t) \geq C_i^{\text{fire}}(t), \quad \pi_i(t) \notin \{\text{compressed}, \text{quarantine}\}.$$

6.4. Transition Commit Operator

Let the transition commit operator be

$$\mathscr{T}_i : x_i(t) \mapsto x_i(t+1),$$

with

$$x_i(t+1) = \mathscr{T}_i\big(x_i(t), \mathscr{S}_i^{\text{in}}(t), \mathscr{V}_t\big).$$

A committed transition must also generate an event

$$q_i(t) \in \mathscr{Q}_{t+1}.$$

7. Event Construction and Causal Ledger

7.1. Event Structure

Define a committed event as

$$q = (e, u, X^-, X^+, S^{\text{in}}, S^{\text{out}}, t, P),$$

where:

- e is event identifier;
- u is actor neurol or cluster identifier;
- X^- and X^+ are pre-state and post-state summaries or hashes;
- S^{in} and S^{out} are incoming and outgoing signal references;
- t is commit time;
- P is the set of causal parent event identifiers.

7.2. Causal Graph

The ledger is represented as a directed acyclic or nearly acyclic graph

$$\mathscr{G}_Q = (\mathscr{Q}, \mathscr{L}),$$

where each edge in $\mathscr{L}$ expresses causal or transformational ancestry.

7.3. Replay Condition

For replayability, the event representation must satisfy a sufficiency condition:

Proposition 7.1 (Replay Sufficiency)**.** *If the initial substrate state, all committed events, and all required deterministic parameters are known, then the local and global substrate state must be reconstructable up to the declared stochastic boundary.*

This proposition becomes a guiding implementation invariant in later volumes.

8. Memory Formalization

8.1. Local Memory Update

Let the local memory update operator be

$$M_i^{\mathrm{loc}}(t+1) = \mathscr{M}_i\big(M_i^{\mathrm{loc}}(t), q_i(t), \mathscr{C}_i(t)\big),$$

where $\mathscr{C}_i(t)$ is local context.

8.2. Memory Importance Score

A committed event is written to durable memory only if its importance exceeds a threshold. Define importance by

$$\mathscr{I}(q_i(t)) = \beta_1 \Delta a_i(t) + \beta_2 \Delta \Xi_i(t) + \beta_3 \mathscr{R}_i^{\mathrm{sem}}(t) + \beta_4 \mathscr{R}_i^{\mathrm{top}}(t),$$

where:

- $\Delta a_i(t)$ is activation change;
- $\Delta \Xi_i(t)$ is viability change;
- $\mathscr{R}_i^{\mathrm{sem}}(t)$ is semantic relevance;
- $\mathscr{R}_i^{\mathrm{top}}(t)$ is topological impact.

Durable write occurs if

$$\mathscr{I}(q_i(t)) \geq \Theta_i^{\mathrm{mem}}.$$

8.3. Memory Consolidation

Let $\mathscr{B}_i(t)$ denote a batch of related memory items. The consolidated memory centroid is

$$\bar{m}_i(t) = \frac{1}{|\mathscr{B}_i(t)|} \sum_{m \in \mathscr{B}_i(t)} \phi(m),$$

where ϕ embeds the memory item into a representational space.

Then consolidation updates compressed memory by

$$M_i^{\mathrm{cmp}}(t+1) = (1-\lambda_c)M_i^{\mathrm{cmp}}(t) + \lambda_c \bar{m}_i(t).$$

8.4. Latent Inhibition

Repeated patterns should become backgrounded. Let the recurrence count of pattern p for neurol i be $c_i(p,t)$. Define inhibition score

$$L_i(p,t) = 1 - \exp(-\gamma_L c_i(p,t)).$$

A higher $L_i(p,t)$ suppresses immediate attention without deleting the pattern.

9. Identity Trajectory Formalization

9.1. Identity as a Dynamical Variable

Let the identity state of neurol i be represented by a vector

$$z_i(t) \in \mathbb{R}^{m_z}.$$

Then identity evolves as

$$z_i(t+1) = \mathscr{Z}_i\big(z_i(t), q_i(t), M_i^{\mathrm{cmp}}(t+1), \mathscr{N}_i^{\mathrm{nbr}}(t)\big),$$

where $\mathscr{N}_i^{\mathrm{nbr}}(t)$ is a neighborhood summary.

9.2. A Linear-Nonlinear Hybrid Update

A simple but expressive form is

$$z_i(t+1) = \rho_z z_i(t) + W_q \phi(q_i(t)) + W_m M_i^{\mathrm{cmp}}(t+1) + W_n v_i(t),$$

where $v_i(t)$ is aggregated neighborhood influence.

9.3. Trajectory Distance

The identity difference between two times may be measured by

$$D_i^{\mathrm{id}}(t_1,t_2) = \|z_i(t_2) - z_i(t_1)\|.$$

This quantity is later useful for detecting maturation, drift, rupture, and transformation significance.

10. Semantic Space and Dynamic Context Field

10.1. Semantic Coordinates

Each neurol or higher structure carries semantic coordinates

$$v_i(t) \in \mathbb{R}^m, \qquad \|v_i(t)\| = 1.$$

10.2. Anchor Set Construction

Let the candidate neighbor set at the same abstraction level be $\mathcal{N}_i^{(q)}(t)$. Then the anchor set is

$$A_i(t) = \mathrm{TopK}_{j \in \mathcal{N}_i^{(q)}(t)} \left\langle v_i(t), v_j(t) \right\rangle.$$

10.3. Context Plane Construction

Given a selected anchor subset $S \subseteq A_i(t)$ with $|S| = k$, define centroid

$$\mu_S = \frac{1}{k} \sum_{j \in S} v_j.$$

Let the matrix

$$M_S = \begin{bmatrix} (v_{j_1} - \mu_S)^\top \\ \vdots \\ (v_{j_k} - \mu_S)^\top \end{bmatrix}$$

be the centered anchor matrix. If the top principal axes of M_S are a_1, a_2, a_3, then the projection of entity i into frame S is

$$\mathrm{proj}_S(i) = \big(\langle v_i - \mu_S, a_1 \rangle, \langle v_i - \mu_S, a_2 \rangle, \langle v_i - \mu_S, a_3 \rangle \big).$$

10.4. Context-Dependent Semantic Distance

For two entities i, j, define

$$d_S(i,j) = 1 - \frac{\langle \mathrm{proj}_S(i), \mathrm{proj}_S(j) \rangle}{\|\mathrm{proj}_S(i)\| \, \|\mathrm{proj}_S(j)\| + \varepsilon}.$$

10.5. Semantic Resonance

We define the semantic resonance between i and j by

$$\mathscr{R}_{ij}^{\mathrm{sem}}(t) = \alpha_A \frac{\big|A_i(t) \cap A_j(t)\big|}{\max(|A_i(t)|, |A_j(t)|)} + \alpha_V \left\langle v_i(t), v_j(t) \right\rangle + \alpha_P \exp\big(-d_\Phi(i,j)\big),$$

where $d_\Phi(i,j)$ is the active-frame distance under a chosen context plane.

10.6. Imagination Mapping

Let B be an entity to project and A_x a context-borrowing entity. The first-order imagination mapping is

$$f^{(1)}(B, A_x, i) = \{B\} \cup \big(A_{A_x} \setminus \{a_i\}\big),$$

where a_i is one anchor removed from A_x's neighborhood. This produces candidate hybrid frames. The full search tree may be explored recursively as in the DCF construction.

11. Topology and Edge Dynamics

11.1. Edge State

Let an edge between i and j be represented by

$$e_{ij}(t) = \big(w_{ij}(t), \ell_{ij}(t), c_{ij}(t), \tau_{ij}(t)\big),$$

where:

- $w_{ij}(t)$ is strength;
- $\ell_{ij}(t)$ is latency;
- $c_{ij}(t)$ is capacity or throughput;
- $\tau_{ij}(t)$ is trust or historical quality.

11.2. Edge Strength Update

A generic Hebbian-resonant update is

$$w_{ij}(t+1) = (1-\lambda_w)w_{ij}(t) + \eta_w \mathscr{C}_{ij}(t) - \gamma_w \mathscr{N}_{ij}(t),$$

where:

- $\mathscr{C}_{ij}(t)$ is a co-activation or cooperative success term;
- $\mathscr{N}_{ij}(t)$ is a noise, failure, or interference term.

11.3. Bridge Discovery

A candidate long-range edge (i, j) is proposed if the bridge score

$$\mathscr{B}_{ij}(t) = \beta_1 \mathscr{R}_{ij}^{\text{sem}}(t) + \beta_2 \mathscr{R}_{ij}^{\text{top}}(t) + \beta_3 \mathscr{U}_{ij}(t) - \beta_4 \text{cost}_{ij}(t)$$

exceeds a threshold Θ^{bridge}.

12. Resource Physics

12.1. Local Resource State

Each neurol has resource stock $R_i(t) \geq 0$. The substrate has reservoir resource

$$R_{\mathscr{V}}(t) \geq 0.$$

12.2. Resource Update

A local resource account updates by

$$R_i(t+1) = R_i(t) + G_i(t) - C_i(t) + U_i^{\text{rec}}(t) - T_i^{\text{ent}}(t),$$

where:

- $G_i(t)$ is allocated gain;
- $C_i(t)$ is direct cost burn;
- $U_i^{\text{rec}}(t)$ is recovered or recycled resource;
- $T_i^{\text{ent}}(t)$ is entropy tax.

12.3. Activation Cost

A general activation cost may be defined as

$$C_i^{\text{fire}}(t) = c_0 + c_1 \|F_i(t)\| + c_2 \left|S_i^{\text{out}}(t)\right| + c_3 \Delta M_i(t),$$

where $\Delta M_i(t)$ is memory write burden.

12.4. Signal Propagation Cost

For a signal routed along path $p = (i_0, i_1, \ldots, i_L)$,

$$C^{\text{sig}}(p) = \sum_{k=0}^{L-1} \left(c_d \ell_{i_k i_{k+1}} + c_w / (w_{i_k i_{k+1}} + \varepsilon) + c_b b_{i_k i_{k+1}} \right),$$

where b_{ij} captures congestion.

12.5. Entropy Tax

Entropy tax can be modeled as

$$T_i^{\text{ent}}(t) = \lambda_R R_i(t) + \lambda_M \left|M_i^{\text{loc}}(t)\right| + \lambda_E \sum_j \mathbb{1}_{e_{ij}(t) \neq 0},$$

which penalizes maintaining large active structures without sufficient stabilizing return.

13. Existence Quantum, Reward, and Cooperation

13.1. Existence Quantum

Let the viability or existence quantum of neurol i be

$$\Xi_i(t) \in \mathbb{R}.$$

Higher values indicate greater viability and continuation capacity.

13.2. Existence Update Rule

We define

$$\Xi_i(t+1) = \Xi_i(t) + \mathscr{W}_i^{\text{imp}}(t) + \mathscr{W}_i^{\text{coop}}(t) - \mathscr{P}_i^{\text{waste}}(t) - \mathscr{P}_i^{\text{destab}}(t),$$

where:

- $\mathscr{W}_i^{\text{imp}}(t)$ is impact reward;
- $\mathscr{W}_i^{\text{coop}}(t)$ is cooperation reward;
- $\mathscr{P}_i^{\text{waste}}(t)$ is waste penalty;
- $\mathscr{P}_i^{\text{destab}}(t)$ is destabilization penalty.

13.3. Impact Reward

A useful impact term is

$$\mathscr{W}_i^{\text{imp}}(t) = \gamma_1 \mathscr{D}_i^{\text{prop}}(t) + \gamma_2 \mathscr{R}_i^{\text{sem}}(t) + \gamma_3 \mathscr{C}_i^{\text{clu}}(t),$$

where:

- $\mathscr{D}_i^{\text{prop}}(t)$ measures downstream propagation depth or breadth;
- $\mathscr{R}_i^{\text{sem}}(t)$ measures semantic contribution;
- $\mathscr{C}_i^{\text{clu}}(t)$ measures cluster stabilization contribution.

13.4. Cooperation Score

Let

$$\Gamma_i(t) = \delta_1 \mathscr{F}_i^{\text{fair}}(t) + \delta_2 \mathscr{S}_i^{\text{stb}}(t) - \delta_3 \mathscr{X}_i^{\text{conf}}(t),$$

where:

- $\mathscr{F}_i^{\text{fair}}(t)$ is fair-exchange contribution;
- $\mathscr{S}_i^{\text{stb}}(t)$ is stabilizing contribution;
- $\mathscr{X}_i^{\text{conf}}(t)$ is conflict or exploit burden.

Then we may set

$$\mathscr{W}_i^{\text{coop}}(t) = \gamma_4 \Gamma_i(t).$$

13.5. Influence Update

Influence evolves by

$$I_i(t+1) = (1-\lambda_I) I_i(t) + \nu_1 \mathscr{W}_i^{\text{imp}}(t) + \nu_2 \Gamma_i(t) - \nu_3 \mathscr{A}_i^{\text{abuse}}(t).$$

14. Attention Allocation

14.1. Attention Priority Score

Let the attention priority for neurol i at time t be

$$A_i^{\text{prio}}(t) = \alpha_1 \mathscr{N}_i^{\text{nov}}(t) + \alpha_2 \mathscr{T}_i^{\text{tens}}(t) + \alpha_3 \mathscr{R}_i^{\text{sem}}(t) + \alpha_4 \mathscr{Q}_i^{\text{risk}}(t) - \alpha_5 L_i(t),$$

where $L_i(t)$ is latent inhibition.

14.2. Attention Allocation Distribution

Given a finite substrate attention budget $A_{\mathscr{V}}(t)$, allocate attention shares by

$$\mathfrak{a}_i(t) = A_{\mathscr{V}}(t) \cdot \frac{\exp(A_i^{\text{prio}}(t)/T_A)}{\sum_j \exp(A_j^{\text{prio}}(t)/T_A)},$$

where T_A is an allocation temperature.

14.3. Effect of Attention

Attention may modulate drive accumulation, memory writing, semantic update fidelity, and resource allocation. For instance,

$$\psi_i(t) \leftarrow \psi_i(t) + \alpha_A \mathfrak{a}_i(t).$$

15. Cluster Emergence and Agent Promotion

15.1. Local Coherence Between Neurols

Define pairwise coherence

$$\mathscr{K}_{ij}(t) = \beta_1 \mathscr{R}_{ij}^{\text{sem}}(t) + \beta_2 \mathscr{R}_{ij}^{\text{sig}}(t) + \beta_3 \mathscr{R}_{ij}^{\text{id}}(t) + \beta_4 w_{ij}(t),$$

where:

- $\mathscr{R}_{ij}^{\text{sig}}$ captures mutual signal success;
- $\mathscr{R}_{ij}^{\text{id}}$ captures identity-trajectory compatibility.

15.2. Cluster Score

For a candidate subset $C \subseteq \mathscr{N}_t$, define cluster score

$$\mathscr{C}(C,t) = \frac{1}{|C|^2} \sum_{i,j \in C} \mathscr{K}_{ij}(t) - \lambda_C \text{boundary}(C,t),$$

where $\text{boundary}(C,t)$ penalizes weak or unstable boundary definition.

A set C becomes a cluster if

$$\mathscr{C}(C,t) \geq \Theta^{\text{cluster}}$$

for a sufficient persistence window.

15.3. Agent Promotion

A cluster C is promoted to agent status if

$$\mathscr{C}(C,t) \geq \Theta^{\text{agent}}, \qquad \Pi(C,t) \geq \Theta^{\text{persist}}, \qquad \Sigma(C,t) \geq \Theta^{\text{self}},$$

where:

- $\Pi(C,t)$ measures temporal persistence;
- $\Sigma(C,t)$ measures self-maintaining capability.

16. ARC Reproduction and Mutation

16.1. Entity Representation for Reproduction

At the reproductive layer, let an entity be represented by

$$E = (G, X, A, K, q, d),$$

where:

- G is genotype or compressed identity core;
- X is execution context;
- A is anchor set;
- K is known-combo or internalized experience set;
- q is abstraction level;
- d is domain label.

16.2. Compression Operator

Compression maps entity E to a seed

$$\mathscr{C}(E) = S = (G, \varepsilon, k),$$

where ε is epigenetic or inheritance mask and k is bootstrap/centrosomal vector.

16.3. Decompression Operator

Given a seed S and substrate entity V, define

$$\mathscr{D}(S, V) = E',$$

with child genotype coordinate-wise defined by

$$G'[m] = \begin{cases} G_S[m], & \varepsilon[m] = 1, \\ \frac{G_S[m]+G_V[m]}{2} + \eta_m, & \varepsilon[m] = 0, \end{cases}$$

where $\eta_m \sim \mathscr{N}(0, \sigma^2_{\text{mut}})$.

16.4. Anchor Inheritance

The child anchor set may be defined by

$$A' = A_V \cup (A_S \setminus A_V)_{\leq k},$$

meaning that the child inherits the receiving substrate's anchor set plus limited novel anchors from the seed side.

16.5. Interaction Types

Let interaction type $\tau \in \{DD, DH, HD, HH\}$. The type determines whether merge, asymmetric reproduction, mirrored reproduction, or nonviable mutual compression occurs.

16.6. Reproduction Eligibility

An entity or cluster is reproduction-eligible if

$$\Xi_i(t) \geq \Theta^{\text{repr}}_{\Xi}, \qquad R_i(t) \geq \Theta^{\text{repr}}_{R},$$

and additional context or compatibility conditions hold.

17. Compression, Dormancy, and Quarantine Operators

17.1. Compression Trigger

Compression occurs if viability remains too low or maintenance cost too high:

$$\Xi_i(t) < \Theta_\Xi^{\mathrm{cmp}} \quad \text{or} \quad R_i(t) < \Theta_R^{\mathrm{cmp}}.$$

Then mode may transition as

$$\pi_i(t+1) = \text{compressed}.$$

17.2. Dormancy Trigger

Dormancy is entered when activity pressure is low but preservation remains justified:

$$a_i(t) < \Theta_i(t), \qquad \Xi_i(t) \geq \Theta_\Xi^{\min}, \qquad \mathfrak{a}_i(t) \ll 1.$$

17.3. Quarantine Trigger

Quarantine occurs when exploit risk or destabilization burden is too high:

$$\mathscr{A}_i^{\mathrm{abuse}}(t) + \mathscr{P}_i^{\mathrm{destab}}(t) \geq \Theta^{\mathrm{quar}}.$$

18. Global Runtime Update Equations

18.1. Canonical Neurol Update

For each active neurol, the canonical local update at time t may be summarized as:

$$\begin{aligned}
F_i(t) &= \sum_{s \in \mathscr{S}_i^{\mathrm{in}}(t)} \omega_s \varepsilon_s g_\theta(y_s, x_i(t)), \\
d_i^{\mathrm{acc}}(t+1) &= \beta_d d_i^{\mathrm{acc}}(t) + \varphi_i(t) + \psi_i(t) - \omega_i(t), \\
a_i(t+1) &= w_d d_i^{\mathrm{acc}}(t+1) + w_f \|F_i(t)\| + w_m m_i^{\mathrm{act}}(t) - w_h h_i(t) - w_\tau \tau_i(t), \\
x_i(t+1) &= \begin{cases} \mathscr{T}_i(x_i(t), \mathscr{S}_i^{\mathrm{in}}(t), \mathscr{V}_t), & a_i(t+1) \geq \Theta_i(t), \\ \mathscr{U}_i(x_i(t), \mathscr{S}_i^{\mathrm{in}}(t), \mathscr{V}_t), & \text{otherwise}, \end{cases}
\end{aligned}$$

where $\mathscr{U}_i$ is the non-firing update operator.

18.2. Post-Commit Global Pass

After local transitions, the system performs:

1. event append to ledger;
2. memory and identity updates;
3. resource deduction and recycling;
4. reward and influence updates;

5. topology adaptation;
6. semantic anchor and context refresh if scheduled;
7. cluster and agent emergence checks;
8. reproduction, dormancy, compression, or quarantine checks.

19. Invariants and Boundedness Conditions

19.1. Non-Negativity of Resource

For all i, t,

$$R_i(t) \geq 0, \qquad R_{\mathscr{V}}(t) \geq 0.$$

Any implementation that permits meaningful negative resource without explicit debt semantics violates the intended substrate physics.

19.2. Finite Event Commitment

At each discrete scheduler step, only finitely many events may be committed:

$$|\mathscr{Q}_{t+1} \setminus \mathscr{Q}_t| < \infty.$$

This ensures bounded processability.

19.3. Existence Preservation

If a neurol or higher structure leaves active mode, it must transition into one of the valid continuation states:

$$\text{active} \to \{\text{latent}, \text{dormant}, \text{compressed}, \text{merged}, \text{quarantine}\}$$

possibly with ledger persistence. Primitive annihilation is not an allowed ontological shortcut.

19.4. Replay Consistency

Given deterministic parameter settings and declared stochastic seeds, replay from checkpoint plus event suffix must reconstruct the same committed event history.

19.5. Bounded Reward Growth

To prevent runaway monopolies, influence and viability growth may be regularized, e.g.

$$\Xi_i(t+1) \leftarrow \Xi_i(t+1) - \lambda_\Xi \Xi_i(t)^2,$$

$$I_i(t+1) \leftarrow \min\{I_i(t+1), I_{\max}\}.$$

These regularizers are parameter choices rather than eternal laws, but some boundedness mechanism is necessary.

20. Recommended Formal Parameter Families

20.1. Core Parameter Groups

The formal system naturally groups parameters into families:

a. activation parameters: $w_d, w_f, w_m, w_h, w_\tau, \Theta_i^{(0)}$;

b. drive parameters: $\beta_d, \alpha_F, \alpha_M, \alpha_K, \alpha_R$;

c. resource parameters: $c_0, c_1, c_2, c_3, \lambda_R, \lambda_M, \lambda_E$;

d. semantic parameters: $\alpha_A, \alpha_V, \alpha_P$;

e. reward parameters: $\gamma_1, \gamma_2, \gamma_3, \gamma_4$;

f. cluster parameters: $\Theta^{\text{cluster}}, \Theta^{\text{agent}}, \lambda_C$;

g. evolutionary parameters: $\sigma_{\text{mut}}, \Theta_\Xi^{\text{repr}}, \Theta_R^{\text{repr}}$.

20.2. Why Parameters Must Remain Explicit

A theory that hides its tunables behind vague prose becomes untestable. A theory that exposes them can be calibrated, compared, and falsified.

21. What Remains for *Chuyen* I and *Chuyen* II

21.1. Why Formalization Is Not the End

Even a complete formal system does not yet tell us how to organize repository layout, runtime communication, interfaces, event buses, or fractal folder architectures. Those are the tasks of the *Chuyen* volumes.

21.2. What *Chuyen* I Must Do

Chuyen I must map the formal system into a practical source architecture with runtime modules, message boundaries, data models, persistence layers, and engineering contracts.

21.3. What *Chuyen* II Must Do

Chuyen II must then refine that architecture into a fractal style so that the nonlinear and recursive nature of the system is mirrored in code organization rather than lost in a flat engineering shell.

22. Conclusion

Dan completes the second great movement of the revised neurol series. It takes the ontological world of *Sinh* and gives it explicit mathematics. The substrate is no longer described only in philosophical language. It now possesses variables, operators, thresholds, score functions, constraints, and update laws.

This formalization does not eliminate future choice. It clarifies where choice lives. It shows what must be computed, what must be preserved, what must remain bounded, and what must later be encoded in software. It also ensures that the later implementation layers do not drift into arbitrary engineering patterns divorced from the theory.

If *Sinh* established what kinds of things are real in the system, *Dan* establishes how those things move.